

Hong Chul Nam

✉ i.am.hongchulnam@gmail.com | 🏠 hongchul-nam.github.io | 🌐 hongchul.nam | 📺 hongchul-nam | 🐦 @HongChulNam

EDUCATION

California Institute of Technology

Visiting Researcher

Pasadena, USA

Oct 2023 – Sept 2024

ETH Zurich

Master of Science in Electrical Engineering and Information Science

Zurich, Switzerland

May 2022 – Present

ETH Zurich

Bachelor of Science in Electrical Engineering and Information Science

Zurich, Switzerland

Sep 2018 – Sep 2022

St. Anselm's Abbey School

High School

Washington DC, USA

Sep 2014 – Sep 2016

EXPERIENCE

AI Researcher

Korean Military Academy

Oct 2024 – Present

Seoul, Republic of Korea

- Supervisor: Sunil Hwang, Prof. Hyun Kwon
- Mandatory military service (1.5 years)
- Building an autonomous surveillance system using real-time object detection with VLM for the Korean army
- Implementing adversarial patches on military clothing to achieve adversarial camouflage against autonomous surveillance systems

Visiting Researcher

Anima's Lab, California Institute of Technology

Oct 2023 – Present

Pasadena, USA

- Funding: Caltech Research Funding
- Supervisor: Dr. Julius Berner, Prof. Anima Anandkumar
- Solved high-dimensional Poisson equations using stochastic reformulation via Walk-on-Spheres (WoS) and neural network in a general framework called Neural Walk-on-Spheres (NWoS)
- Extending self-supervised learning methods (VICReg, MAE, JEPa) to functions with neural operators
- Collaborating with researchers from NVIDIA, University of Chicago, Argonne National Laboratory and Caltech to improve weather modeling with self-supervised learning
- Co-supervising a student on extending Neural Walk-on-Spheres (NWoS) for path planning
- Collaborating with a researcher from Purdue University to extend Neural Walk-on-Spheres (NWoS) to neural operators (WoS-NO)
- Achievements: publication in ICML 2024, submission to ICML 2025

AI Researcher

Alsemy Inc.

Jan 2022 – Dec 2024

Seoul, Republic of Korea

- Supervisor: Chanwoo Park, Dr. Ye Sle Cha, Hyunbo Cho
- Accelerated modeling of variability in transistor devices under process variations using generative models and importance sampling
- Implemented neural operator-based foundation models and reinforcement learning methods for transistor device modeling and circuit optimization
- Created the first function-level anomaly detection neural operator for detecting anomaly device simulations
- Proposed a neural ODE-based compact model for transient self-heating simulation of transistor devices
- Achievements: publications in SISPAD 2023/2024, NeurIPS 2023 Workshop, IEEE Electron Device Letters Submission; full simulation pipeline for variability analysis, DTCO, anomaly detection and self-heating

Research Assistant

SAFARI Lab, ETH Zurich

Sep 2021 – Jul 2023

Zurich, Switzerland

- Supervisor: Konstantinos Kanellopoulos, Prof. Onur Mutlu
- Created a neural network-based page table walk cost estimator to selectively place evicted L2 TLB page tables into L2 data cache
- Introduced Merkle trees in page tables to enforce efficient security check during each intermediate walk of page table walks to verify the integrity of page tables under security attacks
- Contributed to CUDA programming introductory codes for a heterogeneous computing course
- Achievements: publication in MICRO 2023; extension of SniperSim to support neural network; extension of Gem5 and SniperSim to support Merkle trees in page tables

Research Intern

Swiss Center for Electronics and Microtechnology (CSEM)

Sep 2021 – Apr 2022

Zurich, Switzerland

- Supervisor: Dr. Andrea Bonetti
- Developed a low-power MFCC feature extractor using SystemVerilog with UVM verification for a speech signal AI accelerator
- Explored structural changes in hardware as well as neural network architecture for achieving ultra-low power consumption
- Achievements: implementation of MFCC module in SystemVerilog code with verification; optimization of hardware/software implementation of the block

Research Engineer

ASL Lab, ETH Zurich

Sep 2020 – Sep 2021

Zurich, Switzerland

- Funding: Swiss military, ETH Zurich
- Supervisor: David Rohr, Dr. Nicholas RJ Lawrance, Prof. Roland Siegwart
- Developed an autonomous firefighting VTOL drone for wildfire surveillance from scratch
- Achievements: development of VTOL drone with surveillance system

Research Assistant

Vanderver Laboratory, Children's National Hospital

Jun 2015 – Jul 2015

Washington DC, USA

- Funding: Summer internship stipend
- Supervisor: Dr. Asako Takanohashi, Prof. Adeline Vanderver
- Worked on Aicardi Goutieres Syndrome using PCR, cell culture, western blot, and immunohistochemistry approaches
- Achievements: learning cellular biology and all techniques for growing cells and treating drugs

PROJECTS

Learning Pushforward Measure Operator with Guided Flow Matching | *Python, PyTorch* Dec 2024 – Present

- Using a classifier-guidance flow matching model to learn pushforward measure operator along with posterior distribution for learning both likelihood and sampling simultaneously

Equivariant Flow Policy for Robot Learning | *Python, PyTorch*

Dec 2024 – Present

- Supervisor: Prof. SooJean Han
- Incorporating equivariance in flow matching to learn equivariant flow policy for reinforcement learning with applications in robot learning

Continuous Control of Robot Arm | *Python, ROS, C++, PyTorch*

Jan 2023 – Jun 2023

- Supervisor: Bhavya Sukhija, Lenart Treven, Prof. Andreas Krause
- Implemented robot API for RL in the real world for the UR5 robot
- Implemented and tested offline model based RL methods on the robot
- Achievements: implemented software interface to control UR5 robot arm; verified RL methods in the robot

Autonomous Driving Mini-car | *C, TensorFlow*

Sep 2020 – Jan 2021

- Supervisor: Dr. Michele Magno, Prof. Luca Benini
- Developed a real-time autonomous driving model using a Sony microcontroller and an RC car with external camera modules for CNN-based lane detection
- Achievements: demonstrated the highly accurate and functional autonomous driving system with Sony microcontrollers using CNN-base architectures

PUBLICATIONS

- Hrishikesh Viswanath, **Hong Chul Nam**, Julius Berner, Anima Anandkumar, Aniket Bera. "Walk-on-Spheres Meets Neural Operators: PDE Modeling on Irregular Geometries". Submission: Forty-second International Conference on Machine Learning (ICML 2025).
- Premkumar Vincent*, Yeongwoo Nam*, Kyungmin Kim*, **Hong Chul Nam**, Hyunseok Whang, Donghyun Jin, Jongwook Jeon, Chang-Sub Lee, Jihun Park, Ye Sle Cha, Hyunbo Cho. "Large Pre-trained Model Approach for Efficient Design Technology Co-Optimization". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).
- **Hong Chul Nam***, Tae Il Oh*, Ye Sle Cha, Jongwook Jeon, Hyunbo Cho. "FuncFlow: A Generative Neural Operator Using Diffusion Model for Simulation Augmentation". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).

- **Hong Chul Nam**^{*}, Julius Berner^{*}, Anima Anandkumar. "Solving Poisson Equations with Neural Walk-on-Spheres". In: Forty-first International Conference on Machine Learning (ICML 2024).
- Tae Il Oh^{*}, **Hong Chul Nam**^{*}, Chanwoo Park, Hyunbo Cho. "FuncAnoDe: A Function Level Anomaly Detection in Device Simulation". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2024).
- Konstantinos Kanellopoulos, **Hong Chul Nam**, Nisa Bostanci, Rahul Bera, Mohammad Sadrosadati, Rakesh Kumar, Davide Basilio Bartolini, Onur Mutlu. "Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources". In: Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO 2023).
- Chanwoo Park^{*}, **Hong Chul Nam**^{*}, Jihun Park, Jongwook Jeon. "FlowSim: An Invertible Generative Network for Efficient Statistical Analysis under Process Variations". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2023).
- **Hong Chul Nam**, Chanwoo Park. "NPC-NIS: Navigating Semiconductor Process Corners with Neural Importance Sampling". In: NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.

TECHNICAL/LINGUISTIC SKILLS

Linguistics: English, Mandarin, Korean, German

Languages: Java, Python, C/C++, Verilog/SystemVerilog

ML Libraries: PyTorch, TensorFlow, JAX

Computer Simulators: Gem5, SniperSim